Setup: Create the Dataset

```
# Load required libraries
install.packages(c("naniar", "VIM", "mice", "ggplot2", "dplyr"))
library(naniar)
library(VIM)
library(mice)
library(ggplot2)
library(dplyr)

# Create staff dataset
staff <- data.frame(
  staff_id  = 1:8,
  age       = c(22, 25, NA, 30, 35, NA, 45, 50),
  income    = c(30000, NA, 45000, 50000, NA, 70000, 80000, 90000),
  education = c("B", "B", "M", NA, "M", "D", "D", "D")
)

str(staff)
print(staff)
```

Problem 1: Detect and Summarize Missing Values

```
# Total missing values in dataset
sum(is.na(staff))

# Missing values per column
colSums(is.na(staff))

# Percentage missing per column
```

```
colMeans(is.na(staff)) * 100

# Which cells are missing (logical matrix)
is.na(staff)

# Rows with at least one missing value
staff[!complete.cases(staff), ]

# Number of complete rows vs incomplete
sum(complete.cases(staff))
sum(!complete.cases(staff))

# naniar summaries
n_miss(staff)
prop_miss(staff)
miss_var_summary(staff)   # missing count/% per column
miss_case_summary(staff)  # missing count/% per row
```

Problem 2: Visualize Missing Data Patterns

(a) VIM Aggregation Plot

```
aggr_plot <- aggr(staff, col = c('skyblue','red'),
                  numbers = TRUE, sortVars = TRUE,
                  labels = names(staff), cex.axis = 0.7,
                  gap = 3, ylab = c("Missing data histogram", "Pattern"))
```

(b) naniar Visualizations

```
# Full missingness matrix (visual heatmap of NA vs present)
vis_miss(staff)
```

```
# Missing count per variable
gg_miss_var(staff) + labs(title = "Missing Values per Column - Staff Data")

# Missingness pattern combinations
gg_miss_upset(staff)

# Relationship between missingness in age and income
ggplot(staff, aes(x = age, y = income)) +
  geom_miss_point(size = 3) +
  labs(title = "Missingness Pattern: Age vs Income")
```

(c) Matrix Plot (VIM)

```
matrixplot(staff, sortby = "income")
```

(d) Bar plot of % missing (ggplot2, manual)

```
missing_df <- data.frame(
  Column = names(staff),
  Missing_Percent = colMeans(is.na(staff)) * 100
)

ggplot(missing_df, aes(x = reorder(Column, -Missing_Percent), y = Missing_Percent)) +
  geom_bar(stat = "identity", fill = "tomato") +
  labs(title = "Percentage of Missing Values per Column", x = "Column", y = "% Missing") +
  theme_minimal()
```

Problem 3: Handle Missing Values

```
# --- Method 1: Listwise deletion ---
staff_complete <- na.omit(staff)
print(staff_complete)
```

```
# --- Method 2: Mean imputation (age) ---
staff_mean <- staff
staff_mean$age[is.na(staff_mean$age)] <- round(mean(staff_mean$age, na.rm = TRUE), 1)

# --- Method 3: Median imputation (income) ---
staff_mean$income[is.na(staff_mean$income)] <- median(staff_mean$income, na.rm = TRUE)

# --- Method 4: Mode imputation (education - categorical) ---
get_mode <- function(x) {
  ux <- na.omit(unique(x))
  ux[which.max(tabulate(match(x, ux)))]
}
staff_mean$education[is.na(staff_mean$education)] <- get_mode(staff$education)

print(staff_mean)

# --- Method 5: KNN Imputation (VIM) ---
staff_knn <- kNN(staff, variable = c("age","income","education"), k = 3)
print(staff_knn[, 1:4])

# --- Method 6: Multiple Imputation using mice (numeric only) ---
imp_model <- mice(staff[, c("age","income")], method = "pmm", m = 5, seed = 123)
staff_mice <- complete(imp_model)
print(staff_mice)
summary(imp_model)
```

Problem 4: Exploratory Plots on Cleaned Data

```
df <- staff_mean   # use mean/median/mode-imputed dataset

# 1. Histogram - Age distribution
ggplot(df, aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "steelblue", color = "black") +
  labs(title = "Age Distribution", x = "Age", y = "Frequency")

# 2. Boxplot - Income by education level
ggplot(df, aes(x = education, y = income, fill = education)) +
  geom_boxplot() +
  labs(title = "Income by Education Level") +
  theme(legend.position = "none")

# 3. Scatter plot - Age vs Income
ggplot(df, aes(x = age, y = income, color = education)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
  labs(title = "Age vs Income (colored by Education)")

# 4. Bar plot - Count by education
ggplot(df, aes(x = education, fill = education)) +
  geom_bar() +
  labs(title = "Staff Count by Education Level") +
  theme(legend.position = "none")

# 5. Density plot - Income
ggplot(df, aes(x = income)) +
```

```
  geom_density(fill = "orange", alpha = 0.5) +
  labs(title = "Density Plot of Income")

# 6. Pie chart - Education proportion
library(dplyr)
library(ggplot2)
edu_counts <- staff_mean %>%
  filter(!is.na(education)) %>%
  dplyr::count(education) %>%
  mutate(percentage = round(100 * n / sum(n), 1))
print(edu_counts)
ggplot(edu_counts, aes(x = "", y = n, fill = education)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar("y") +
  geom_text(aes(label = paste0(n, " (", percentage, "%)")),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Education Level Proportion", fill = "Education") +
  theme_void()
```

Problem 5: Before vs After Imputation Comparison

```
# Use explicit, unambiguous dataset names
staff_before <- staff          # original data (has NA)
staff_after  <- staff_mean      # mean-imputed data (from earlier step)

# Define COMMON breakpoints so both histograms use the same bins
# (this makes "before vs after" visually comparable)
age_breaks <- seq(20, 55, by = 5)

par(mfrow = c(1, 2))
```

```
hist(staff_before$age,
    main = "Age - Before Imputation",
    col = "salmon",
    xlab = "Age",
    breaks = age_breaks,
    xlim = c(20, 55))

hist(staff_after$age,
    main = "Age - After Mean Imputation",
    col = "lightgreen",
    xlab = "Age",
    breaks = age_breaks,
    xlim = c(20, 55))

par(mfrow = c(1, 1))
# Boxplot comparison for income
staff_before <- staff       # original data (has NA in income)
staff_after  <- staff_mean   # mean/median-imputed data

boxplot(staff_before$income, staff_after$income,
       names = c("Before (NA present)", "After (Imputed)"),
       col = c("salmon", "lightgreen"),
       main = "Income Distribution: Before vs After Imputation",
       ylab = "Income")
```